Allais's paradox

Resnik describes this paradox on pp. 103 – 105, but you can just read my description of it here.

This paradox is named for the Nobel Prize winning economist Maurice Allais (1911 – 2010). You have already seen part of this paradox in the form of the guaranteed payoff paradox. Now consider two decisions. This is the first:

	P = .01 state 1	P = .10 state 2	P = .89 state 3
lottery *a*	$1,000,000	$1,000,000	$1,000,000
lottery *b*	$0	$5,000,000	$1,000,000

This is the second:

	state 1	state 2
lottery *c*	$1,000,000 (P = .11)	$0 (P = .89)
lottery *d*	$5,000,000 (P = .10)	$0 (P = .90)

Which lottery would you choose in the first case (between *a* and *b*)? And which would you choose in the second (between *c* and *d*)?

There is no right or wrong answer, but let's proceed with the assumption that most people will choose *a* and *d*. (And we could add more zeros to the $5M and the $1M outcomes. Doing so won't change the issue being investigated here. $5M, $1M, and $0 are just the values that were used in the first published description of the paradox in English in 1974.[1])

The paradox appears to illustrate a problem with the consistency of people's preferences (which are represented by the utility values that they assign to the outcomes). So, we are interested in

[1] Slovic, P., & A. Tversky (1974). "Who Accepts Savage's Axiom?," *Behavioral Science*, 19, 368-373.

the expected utility of each option, not the expected value. If we rank the outcomes from most preferred to least preferred, we have this list:

outcomes	utility values
$5,000,000	1
$1,000,000	u(1M)
$0	0

We don't need to determine a specific utility value for the $1,000,000 outcome, and so we will just refer to it as "u(1M)" (that is, "the utility for $1,000,000").[2] The paradox arises because, whatever u(1M) might be, either

(*i*) EU(lottery *a*) > EU(lottery *b*), and EU(lottery *c*) > EU(lottery *d*)

or

(*ii*) EU(lottery *b*) > EU(lottery *a*), and EU(lottery *d*) > EU(lottery *c*)

Thus, if someone prefers lottery *a* to lottery *b*, then this person should also prefer lottery *c* to lottery *d*. Alternatively, if someone prefers lottery *b* to lottery *a*, then this person should also prefer lottery *d* to lottery *c*.

To prove this, we calculate the expected utility for each option and then find the difference between EU(*a*) and EU(*b*) and EU(*c*) and EU(*d*).

EU(*a*) = (.01)(u(1M)) + (.10)(u(1M)) + (.89)(u(1M)) = (.01 + .10 + .89)(u(1M)) = (1)(u(1M)) = u(1M)

EU(*b*) = (.01)(0) + (.10)(1) + (.89)(u(1M)) = 0 + .10 + (.89)(u(1M)) = .10 + (.89)(u(1M))

EU(*a*) - EU(*b*) = u(1M) - .10 - (.89)(u(1M)) = (1 - .89)(u(1M)) - .10 = **(.11)(u(1M)) - .10**

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[2] Although, as we found when looking at the guaranteed payoff paradox, if a decision-maker chooses *a* over *b*, the utility value for $1,000,000 has to be 1.0 also, or extremely close to it. But that doesn't matter for this either.

EU(*c*) = (.11)(u(1M)) + (.89)(0) = (.11)(u(1M)) + 0 = (.11)(u(1M))

EU(*d*) = (.10)(1) + (.90)(0) = .10 + 0 = .10

EU(*c*) - EU(*d*) = **(.11)(u(1M)) - .10**

Thus, whatever u(1M) happens to be for a person, it has to be the case that either EU(*a*) is greater than EU(*b*) and EU(*c*) is greater than EU(*d*) or the reverse.[3]

Another way to see this is to re-write the second decision table this way:

	P = .01 **state 1**	**P = .10** **state 2**	**P = .89** **state 3**
lottery c	$1,000,000	$1,000,000	$0
lottery d	$0	$5,000,000	$0

and then compare it to the first decision table.

	P = .01 **state 1**	**P = .10** **state 2**	**P = .89** **state 3**
lottery *a*	$1,000,000	$1,000,000	$1,000,000
lottery *b*	$0	$5,000,000	$1,000,000

If we just ignore state 3, which is the same for both outcomes in both decisions, then we see that the decision between *c* and *d* is the same as the decision between *a* and *b*. So, it would seem that everyone should choose either lottery *a* and lottery *c* or lottery *b* and lottery *d*.

[3] If you want to think about the calculations with precise values, then consider u(1M) = 1. This will give us the following. EU(*a*) - EU(*b*) = EU(*c*) - EU(*d*) = .11 - .10 = .01, which means that the expected utilities for *a* and *c* are both (slightly) larger than the expected utilities for *b* and *d*.

Alternatively, if u(1M) = .50, then EU(*a*) - EU(*b*) = EU(*c*) - EU(*d*) = (.11)(.50) - .10 = .055 - .10 = -.045, which means that the expected utilities for *b* and *d* are larger than the expected utilities for *a* and *c*.

Possible solutions to the paradox

Here are two possible solutions to the paradox.

First, it might be that people who choose *a* and *d* misunderstand the problem until they see the second decision table re-written. But once they see the similarity between the two decision problems, they will choose lotteries *a* and *c* or lotteries *b* and *d*.

Second, some people might judge the 1 percent chance at $0 in lottery *b* differently than the 1 percent chance at $0 in lottery *d* (which is really just part of the 90 percent chance of getting $0). Choosing lottery *b* and getting $0 will be associated with the thought that "I could have had a guaranteed $1,000,000!" and a lot of disappointment. Choosing lottery *d* and getting $0 would not be nearly as disappointing. If this is correct, then the two decision problems aren't comparable because they don't have the same least preferred outcomes. Therefore, in addition to the least preferred outcomes being different, relative to the most and least preferred outcomes, u(1M) will also be different in the two decision problems.

Decision 1

outcomes	utility values
$5,000,000	1
$1,000,000	u(1M)
$0 + disappointment	0

Decision 2

outcomes	utility values
$5,000,000	1
$1,000,000	u(1M)
$0	0